Lecture 12: Mesh Current Analysis

EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current, mesh current 1 voltage supermesh

Review: node voltage analysis

The procedure

- Choose the ground; label the node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T \mathbf{e}$.
- Write KCL at every node except the ground.
- Matrix form: $Y_n \mathbf{e} = \mathbf{i}_n$.

Review: node voltage analysis

The procedure

- Choose the ground; label the node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T \mathbf{e}$.
- Write KCL at every node except the ground.
- Matrix form: $Y_n \mathbf{e} = \mathbf{i}_n$.

Voltage sources

- From the ground to node k: e_k is known.
- Between two nodes: a supernode.
- A dependent source: one more constraint.

Review: node voltage analysis

The procedure

- Choose the ground; label the node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T\mathbf{e}$.
- Write KCL at every node except the ground.
- Matrix form: $Y_n\mathbf{e} = \mathbf{i}_n$.

Voltage sources

- From the ground to node k: e_k is known.
- Between two nodes: a supernode.
- A dependent source: one more constraint.

Node voltages make KVL automatic, and we write KCL.

Review: node voltage analysis

The procedure

- Choose the ground; label the node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T\mathbf{e}$.
- Write KCL at every node except the ground.
- Matrix form: $Y_n\mathbf{e} = \mathbf{i}_n$.

Voltage sources

- From the ground to node k: e_k is known.
- Between two nodes: a supernode.
- A dependent source: one more constraint.

Node voltages make KVL automatic, and we write KCL.

Can we do the opposite?

Motivation

Node voltages

- Unknowns: the $n-1$ node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T\mathbf{e}$.
- We write KCL: $n-1$ equations.

Motivation

Node voltages

- Unknowns: the $n-1$ node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T\mathbf{e}$.
- We write KCL: $n-1$ equations.

Mesh currents

- Unknowns: the $\ell = b - n + 1$ mesh currents $\mathbf{i}_M$.

Motivation

Node voltages

- Unknowns: the $n-1$ node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T\mathbf{e}$.
- We write KCL: $n-1$ equations.

Mesh currents

- Unknowns: the $\ell = b - n + 1$ mesh currents $\mathbf{i}_M$.
- KCL holds automatically: $\mathbf{i} = M^T\mathbf{i}_M$ (previously).

Motivation

Node voltages

- Unknowns: the $n-1$ node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T\mathbf{e}$.
- We write KCL: $n-1$ equations.

Mesh currents

- Unknowns: the $\ell = b - n + 1$ mesh currents $\mathbf{i}_M$.
- KCL holds automatically: $\mathbf{i} = M^T\mathbf{i}_M$ (previously).
- We write KVL: ℓ equations.

Motivation

Node voltages

- Unknowns: the $n-1$ node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T\mathbf{e}$.
- We write KCL: $n-1$ equations.

Mesh currents

- Unknowns: the $\ell = b - n + 1$ mesh currents $\mathbf{i}_M$.
- KCL holds automatically: $\mathbf{i} = M^T\mathbf{i}_M$ (previously).
- We write KVL: ℓ equations.
- Meshes need a planar circuit.

Motivation

Node voltages

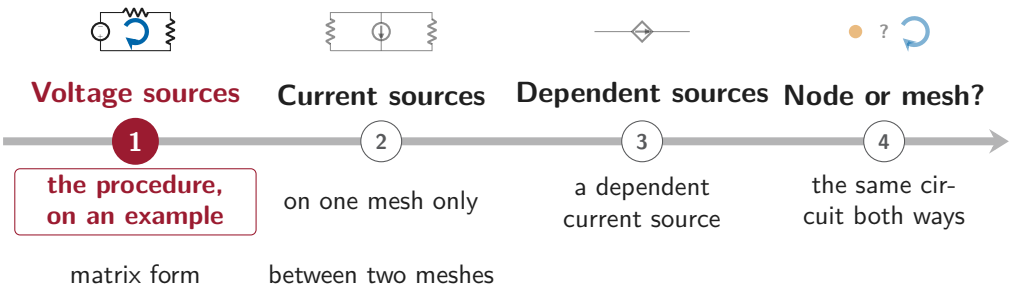
- Unknowns: the $n-1$ node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T\mathbf{e}$.
- We write KCL: $n-1$ equations.

Mesh currents

- Unknowns: the $\ell = b - n + 1$ mesh currents $\mathbf{i}_M$.
- KCL holds automatically: $\mathbf{i} = M^T\mathbf{i}_M$ (previously).
- We write KVL: ℓ equations.
- Meshes need a planar circuit.

Mesh current analysis follows the same plan, with KCL and KVL swapped.

Roadmap

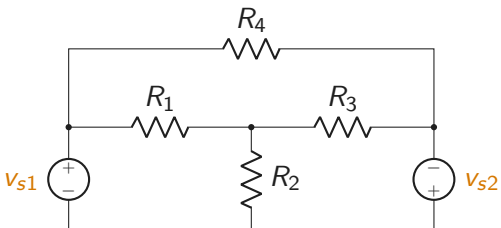


Each station adds a new kind of source; the last one compares the two methods.

Example 12.1: mesh current analysis step by step

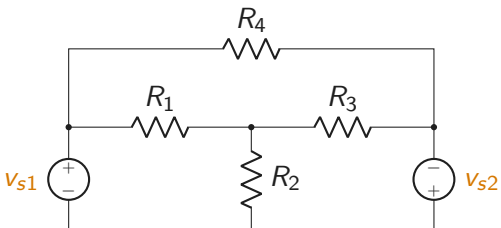
For now: the circuit has only resistors and independent voltage sources.

Question. How many unknowns does this circuit have?



Example 12.1: mesh current analysis step by step

For now: the circuit has only resistors and independent voltage sources.

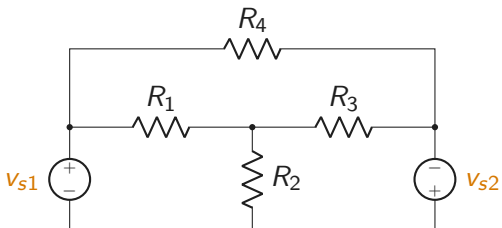


Question. How many unknowns does this circuit have?

$b = 6$ branches: 12 branch voltages and currents.

Example 12.1: mesh current analysis step by step

For now: the circuit has only resistors and independent voltage sources.



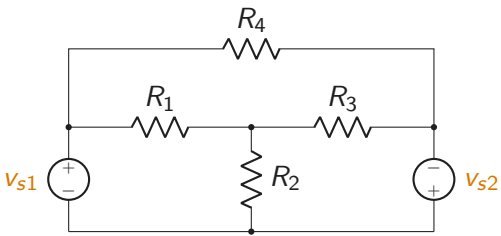
Question. How many unknowns does this circuit have?

$b = 6$ branches: 12 branch voltages and currents.

Mesh current analysis needs only 3 of them.

Example 12.1: mesh current analysis step by step

For now: the circuit has only resistors and independent voltage sources.



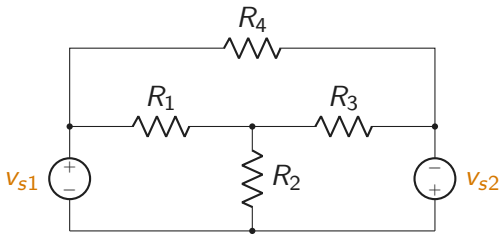
Question. How many unknowns does this circuit have?

$b = 6$ branches: 12 branch voltages and currents.

Mesh current analysis needs only 3 of them.

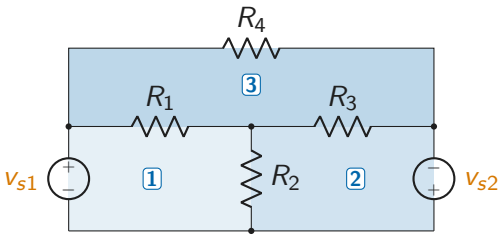
We find them in four steps, on this circuit.

Step 1: find the inner meshes



The circuit is planar: it is drawn with no crossing branches.

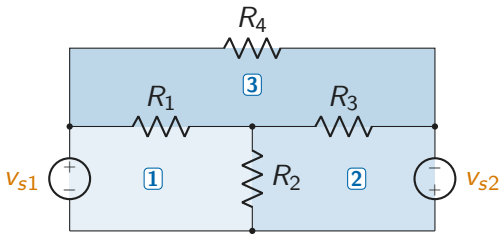
Step 1: find the inner meshes



The circuit is planar: it is drawn with no crossing branches.

Its inner meshes are the windows of the drawing.

Step 1: find the inner meshes

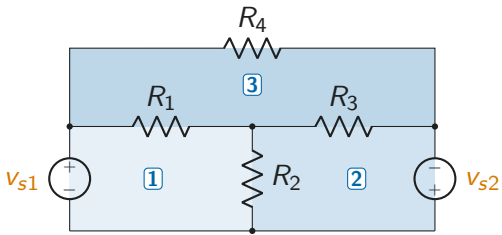


The circuit is planar: it is drawn with no crossing branches.

Its inner meshes are the windows of the drawing.

Here there are three:
$\ell = b - n + 1 = 6 - 4 + 1 = 3$.

Step 1: find the inner meshes



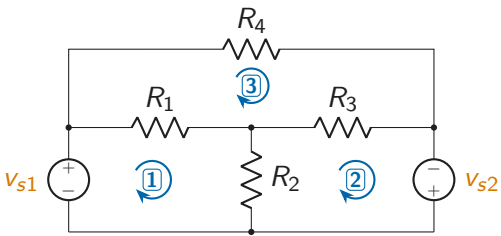
The circuit is planar: it is drawn with no crossing branches.

Its inner meshes are the windows of the drawing.

Here there are three:
$\ell = b - n + 1 = 6 - 4 + 1 = 3$.

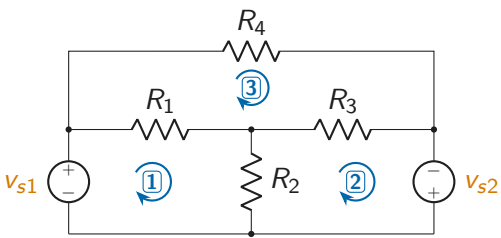
Mesh current analysis has one unknown per inner mesh.

Step 2: assign the mesh currents



Each inner mesh gets a mesh current, clockwise: i_{M1}, i_{M2}, i_{M3}.

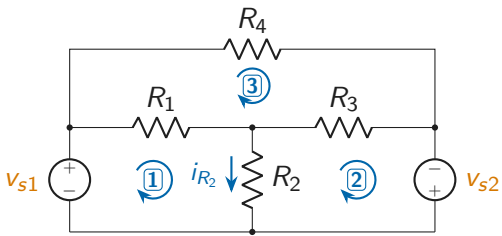
Step 2: assign the mesh currents



Each inner mesh gets a mesh current, clockwise: i_{M1}, i_{M2}, i_{M3}.

These are the unknowns.

Step 2: assign the mesh currents



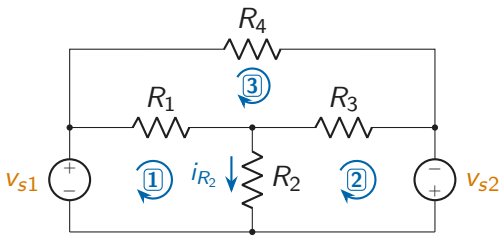
Each inner mesh gets a mesh current, clockwise: i_{M1}, i_{M2}, i_{M3}.

These are the unknowns.

Every branch current is a combination of them:

R_2 carries i_{M1} down and i_{M2} up: $i_{R_2} = i_{M1} - i_{M2}$.

Step 2: assign the mesh currents



Each inner mesh gets a mesh current, clockwise: i_{M1}, i_{M2}, i_{M3}.

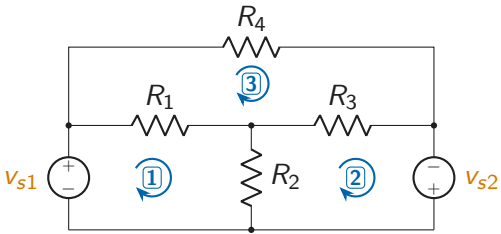
These are the unknowns.

Every branch current is a combination of them:

R_2 carries i_{M1} down and i_{M2} up: $i_{R_2} = i_{M1} - i_{M2}$.

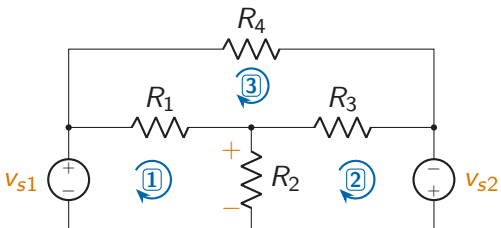
A mesh current enters and leaves every node it passes, so KCL holds automatically.

Step 3: write the resistor voltages with mesh currents



Ohm's law gives each resistor voltage from its mesh currents.

Step 3: write the resistor voltages with mesh currents

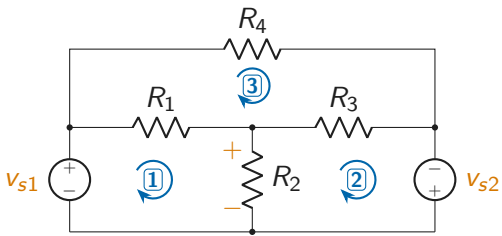


Ohm's law gives each resistor voltage from its mesh currents.

The voltage of R_2 along mesh 1 (top to bottom):

$v = R_2(i_{M1} - i_{M2})$

Step 3: write the resistor voltages with mesh currents



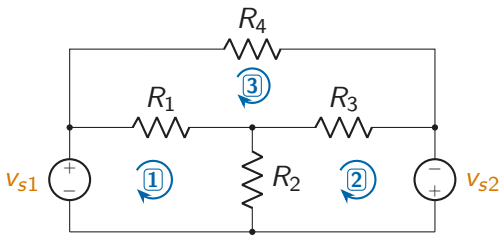
Ohm's law gives each resistor voltage from its mesh currents.

The voltage of R_2 along mesh 1 (top to bottom):

$v = R_2(i_{M1} - i_{M2})$

R_4 is on mesh 3 only: $v = R_4\, i_{M3}$

Step 3: write the resistor voltages with mesh currents



Ohm's law gives each resistor voltage from its mesh currents.

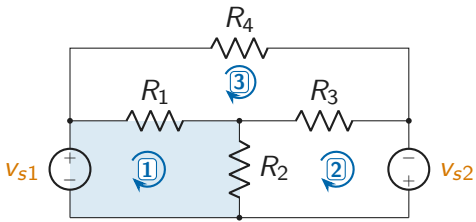
The voltage of R_2 along mesh 1 (top to bottom):

$v = R_2(i_{M1} - i_{M2})$

R_4 is on mesh 3 only: $v = R_4\, i_{M3}$

The rule: R times (own mesh current minus the other one).

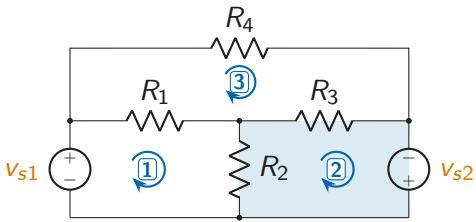
Step 4: KVL around every inner mesh



Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

Step 4: KVL around every inner mesh



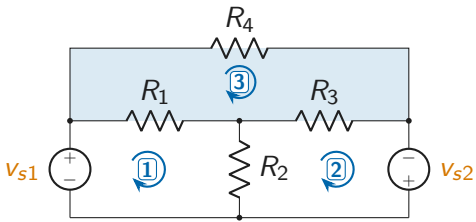
Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

Mesh 2:

$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

Step 4: KVL around every inner mesh



Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

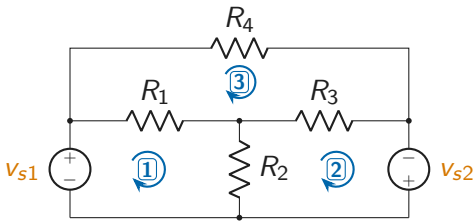
Mesh 2:

$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

Mesh 3:

$$R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0$$

Step 4: KVL around every inner mesh



Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

Mesh 2:

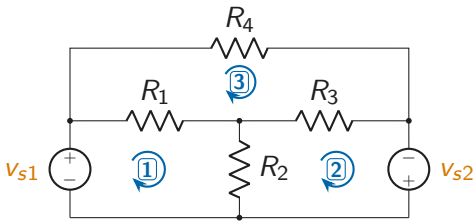
$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

Mesh 3:

$$R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0$$

A source met from $-$ to $+$ enters with a minus sign.

Step 4: KVL around every inner mesh



Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

Mesh 2:

$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

Mesh 3:

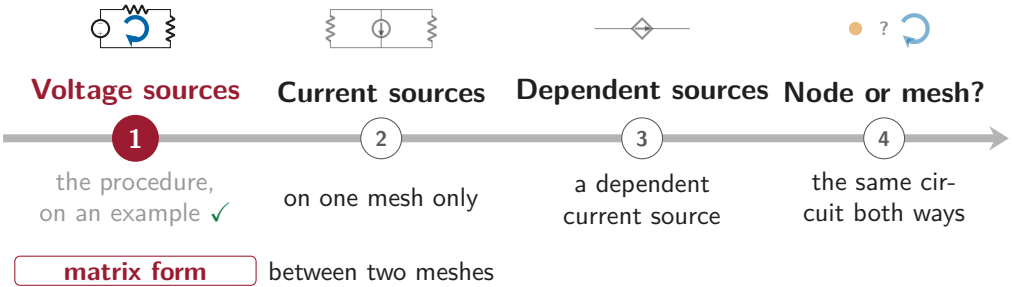
$$R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0$$

A source met from $-$ to $+$ enters with a minus sign.

Three equations, three unknowns.

Then Steps 2 and 3 give every branch current and voltage.

Roadmap



Each station adds a new kind of source; the last one compares the two methods.

Example 12.1 in matrix form

Collect the terms of each mesh:

$$\begin{aligned}(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\ -R_2\, i_{M1} + (R_2 + R_3)\, i_{M2} - R_3\, i_{M3} &= v_{s2} \\ -R_1\, i_{M1} - R_3\, i_{M2} + (R_1 + R_3 + R_4)\, i_{M3} &= 0\end{aligned}$$

Question. Write the three equations as one matrix equation.

Example 12.1 in matrix form

Collect the terms of each mesh:

$$
\begin{aligned}
(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\
-R_2\, i_{M1} + (R_2 + R_3)\, i_{M2} - R_3\, i_{M3} &= v_{s2} \\
-R_1\, i_{M1} - R_3\, i_{M2} + (R_1 + R_3 + R_4)\, i_{M3} &= 0
\end{aligned}
$$

$$
\underbrace{\begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}}_{Z_M}
\underbrace{\begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}}_{\mathbf{i}_M}
=
\underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}
$$

Example 12.1 in matrix form

Collect the terms of each mesh:

$$
\begin{aligned}
(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\
-R_2\, i_{M1} + (R_2 + R_3)\, i_{M2} - R_3\, i_{M3} &= v_{s2} \\
-R_1\, i_{M1} - R_3\, i_{M2} + (R_1 + R_3 + R_4)\, i_{M3} &= 0
\end{aligned}
$$

$$
\underbrace{\begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}}_{Z_M}
\underbrace{\begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}}_{\mathbf{i}_M}
=
\underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}
$$

Z_M is the mesh impedance matrix: $Z_M \mathbf{i}_M = \mathbf{v}_M$.

Example 12.1 in matrix form

Collect the terms of each mesh:

$$
\begin{aligned}
(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\
-R_2\, i_{M1} + (R_2 + R_3)\, i_{M2} - R_3\, i_{M3} &= v_{s2} \\
-R_1\, i_{M1} - R_3\, i_{M2} + (R_1 + R_3 + R_4)\, i_{M3} &= 0
\end{aligned}
$$

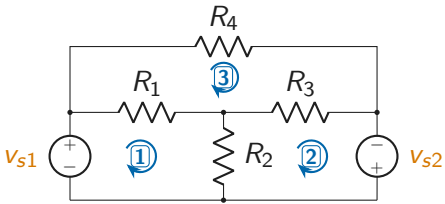
$$
\underbrace{\begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}}_{Z_M}
\underbrace{\begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}}_{\mathbf{i}_M}
=
\underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}
$$

Z_M is the mesh impedance matrix: $Z_M \mathbf{i}_M = \mathbf{v}_M$.

$\mathbf{v}_M$: the source voltages rising along each mesh.

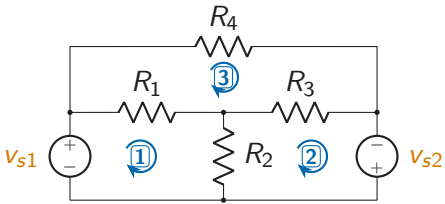
Writing Z_M directly from the circuit

Question. Can we read Z_M from the circuit, without writing KVL?



$$\underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit

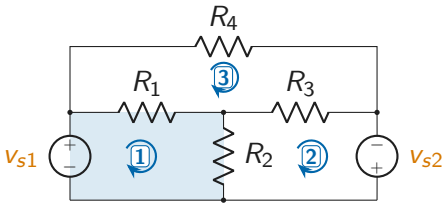


Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

$$\underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit

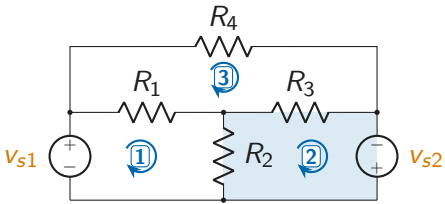


Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

$$\underbrace{\begin{bmatrix} R_1 + R_2 & & \\ & & \\ & & \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit

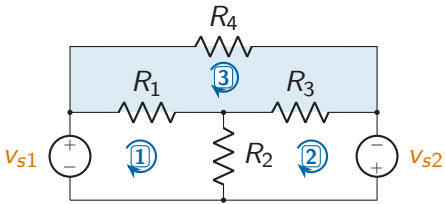


Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

$$\underbrace{\begin{bmatrix} R_1 + R_2 & & \\ & R_2 + R_3 & \\ & & \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit

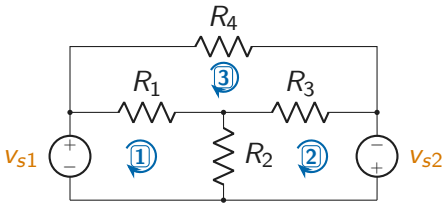


Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

$$\underbrace{\begin{bmatrix} R_1 + R_2 & & \\ & R_2 + R_3 & \\ & & R_1 + R_3 + R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



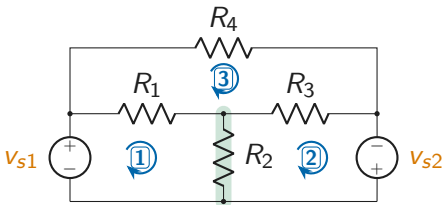
Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

$$\underbrace{\begin{bmatrix} R_1 + R_2 & & \\ & R_2 + R_3 & \\ & & R_1 + R_3 + R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



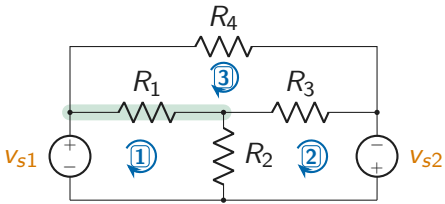
Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

$$\underbrace{\begin{bmatrix} R_1 + R_2 & -R_2 & \\ -R_2 & R_2 + R_3 & \\ & & R_1 + R_3 + R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



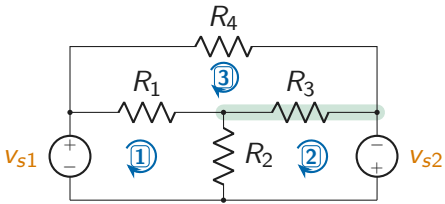
Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

$$\underbrace{\begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & \\ -R_1 & & R_1 + R_3 + R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



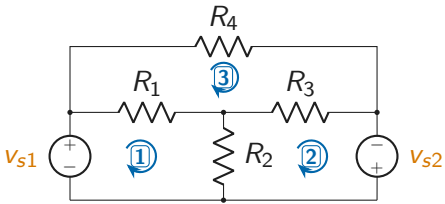
Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



Question. Can we read Z_M from the circuit, without writing KVL?

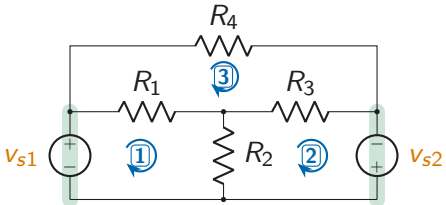
Diagonal (k, k): the sum of the resistances around mesh k.

Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

Right side k: the sum of the source voltages rising (met from $-$ to $+$) along mesh k.

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



Question. Can we read Z_M from the circuit, without writing KVL?

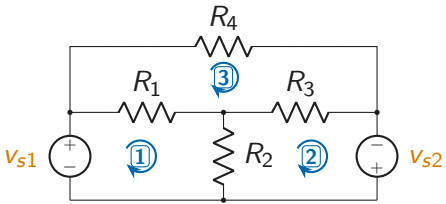
Diagonal (k, k): the sum of the resistances around mesh k.

Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

Right side k: the sum of the source voltages rising (met from $-$ to $+$) along mesh k.

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

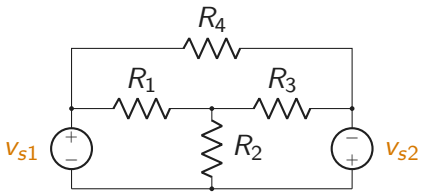
Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

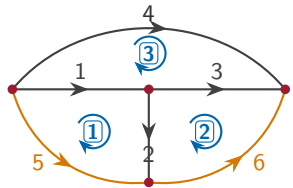
Right side k: the sum of the source voltages rising (met from $-$ to $+$) along mesh k.

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}$$

The same Z_M as from KVL ✓; it is symmetric: entry (k, j) equals entry (j, k).

Z_M from the graph: the mesh matrix

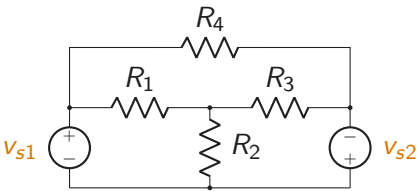


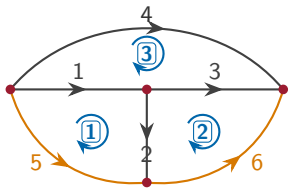


Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

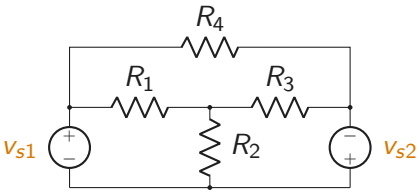
Question. Write the mesh matrix M.

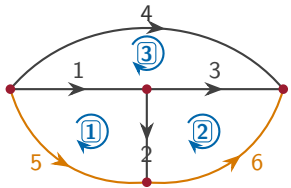
One row for each inner mesh.

Entry $+1$: the branch goes along the mesh; -1: against it.

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

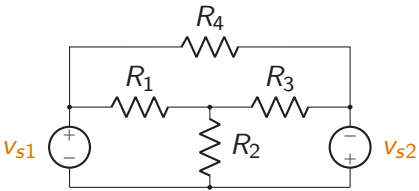
One row for each inner mesh.

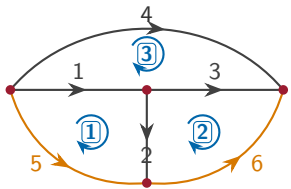
Entry $+1$: the branch goes along the mesh; -1: against it.

Row k: which branches are on mesh k?

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

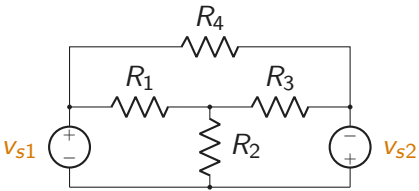
One row for each inner mesh.

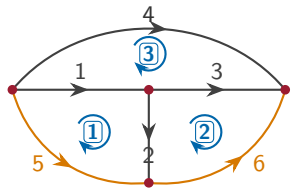
Entry $+1$: the branch goes along the mesh; -1: against it.

Row k: which branches are on mesh k?

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

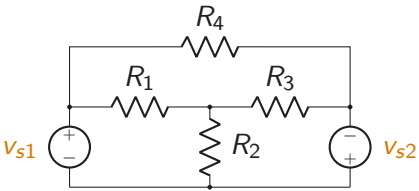
One row for each inner mesh.

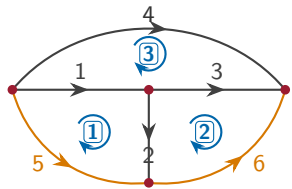
Entry $+1$: the branch goes along the mesh; -1: against it.

Row k: which branches are on mesh k?

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

One row for each inner mesh.

Entry $+1$: the branch goes along the mesh; -1: against it.

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccc|cc} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \end{array}\right] \end{array}$$

M_R: resistors M_S: sources

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL: M^T splits into rows:

$$\mathbf{i} = M^T\mathbf{i}_M : \qquad \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

b_R resistor rows: $\mathbf{i}_R = M_R^T\mathbf{i}_M$

b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL: M^T splits into rows:

$$\mathbf{i} = M^T\mathbf{i}_M : \qquad \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

b_R resistor rows: $\mathbf{i}_R = M_R^T\mathbf{i}_M$

b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

Only the resistor rows are needed: the source currents follow afterwards.

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL: M^T splits into rows:

$$\mathbf{i} = M^T\mathbf{i}_M: \qquad \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

b_R resistor rows: $\mathbf{i}_R = M_R^T\mathbf{i}_M$

b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

Only the resistor rows are needed: the source currents follow afterwards.

Resistors: $\mathbf{v}_R = R\,\mathbf{i}_R$, $R = \mathrm{diag}(R_1, \ldots, R_{b_R})$, one resistance per resistor.

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

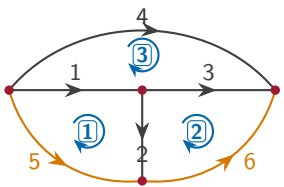
$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

$$Z_M^T = M_R\, R^T M_R^T = M_R\, R\, M_R^T = Z_M, \text{ since } R \text{ is diagonal.}$$

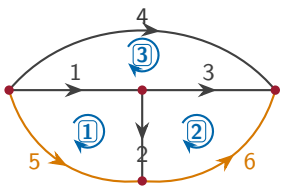
So Z_M is always symmetric for resistors and voltage sources.

Z_M from the graph: Example 12.1, the resistor currents

M_R^T: the resistor columns 1–4 of M, written as rows.



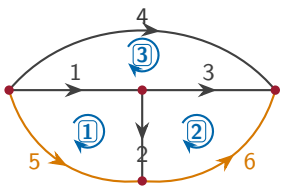
Z_M from the graph: Example 12.1, the resistor currents



M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}$$

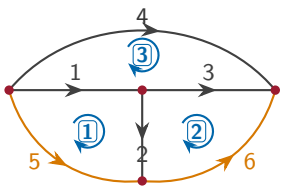
Z_M from the graph: Example 12.1, the resistor currents



M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Z_M from the graph: Example 12.1, the resistor currents

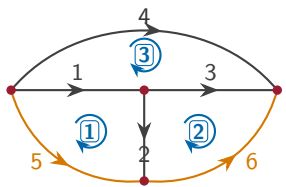


M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Each resistor current: the mesh currents along its arrow minus those against it.

Z_M from the graph: Example 12.1, the resistor currents



M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Each resistor current: the mesh currents along its arrow minus those against it.

Then Ohm's law: $\mathbf{v}_R = R\,\mathbf{i}_R$, $v_{R_k} = R_k i_{R_k}$.

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = - \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = -\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix} = \mathbf{v}_M \quad \checkmark$$

Z_M from the graph: Example 12.1, the matrix

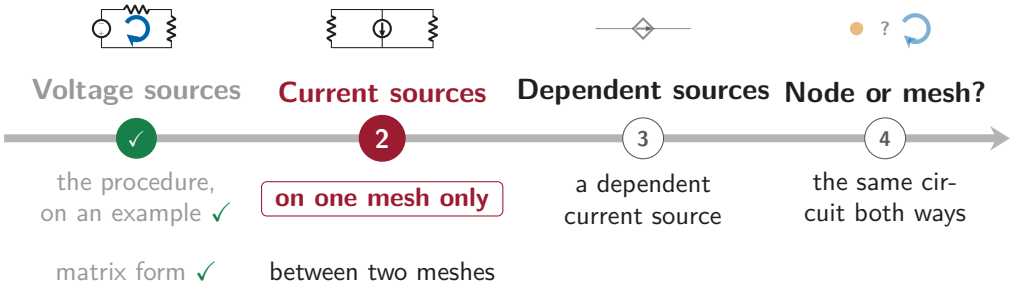
$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = - \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix} = \mathbf{v}_M \quad \checkmark$$

The three rules read this product off, entry by entry.

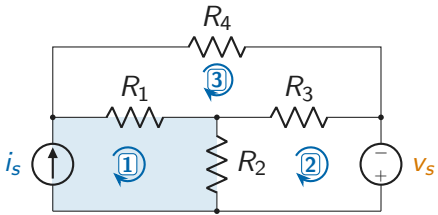
Roadmap



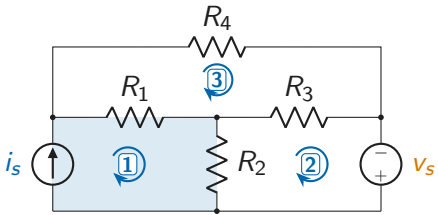
Each station adds a new kind of source; the last one compares the two methods.

Example 12.2: a current source on one mesh only

i_s is on mesh 1 only: $i_{M1} = i_s$ is known.



Example 12.2: a current source on one mesh only

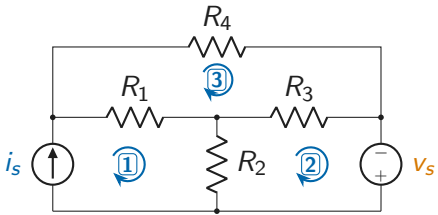


i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

The unknowns are i_{M2}, i_{M3}.

Example 12.2: a current source on one mesh only



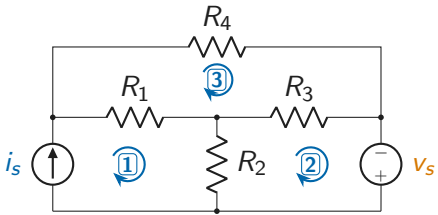
i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

The unknowns are i_{M2}, i_{M3}.

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Example 12.2: a current source on one mesh only



i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

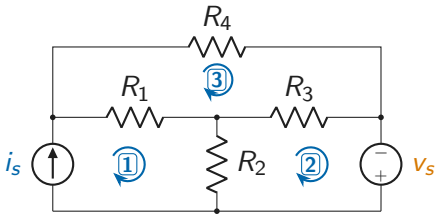
The unknowns are i_{M2}, i_{M3}.

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

Example 12.2: a current source on one mesh only



i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

The unknowns are i_{M2}, i_{M3}.

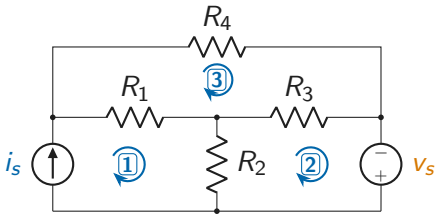
Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

Example 12.2: a current source on one mesh only



i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

The unknowns are i_{M2}, i_{M3}.

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

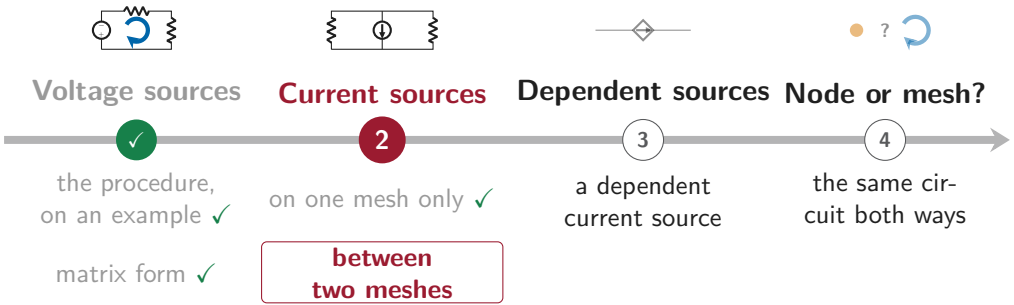
Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

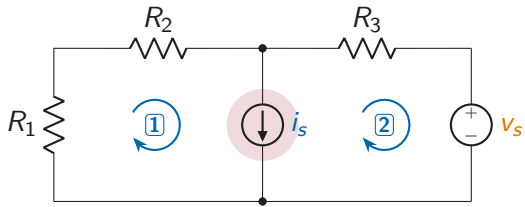
The known i_s moves to the right side.

Roadmap



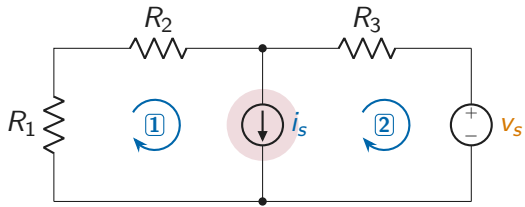
Each station adds a new kind of source; the last one compares the two methods.

Example 12.3: a current source between two meshes



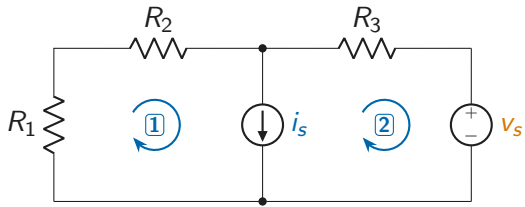
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?

Example 12.3: a current source between two meshes



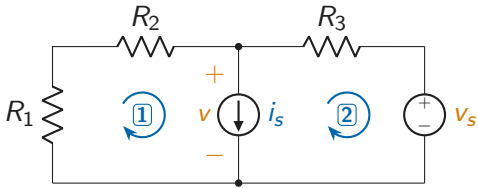
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?
- It cannot be written with i_{M1} and i_{M2}: the law of a current source fixes its current, not its voltage.

Example 12.3: a current source between two meshes



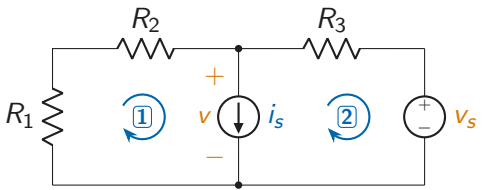
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?
- It cannot be written with i_{M1} and i_{M2}: the law of a current source fixes its current, not its voltage.
- **Modified mesh current analysis** offers two ways out: Method I (its voltage is a new unknown) and Method II (a supermesh).

Method I: the voltage as an unknown



The voltage v of i_s is one more unknown.

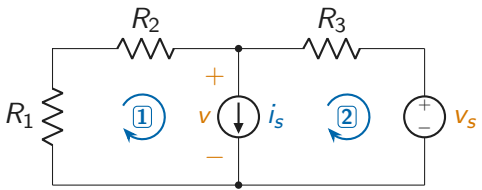
Method I: the voltage as an unknown



Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

The voltage v of i_s is one more unknown.

Method I: the voltage as an unknown

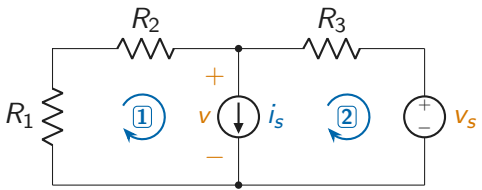


Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

The voltage v of i_s is one more unknown.

Method I: the voltage as an unknown



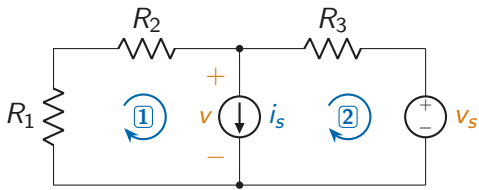
Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

The voltage v of i_s is one more unknown.

Method I: the voltage as an unknown



Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

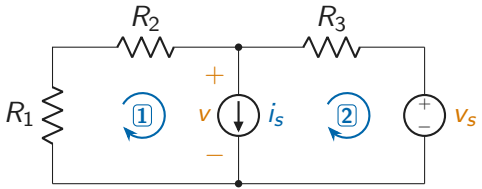
Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

The voltage v of i_s is one more unknown.

The constraint: i_s carries i_{M1} down and i_{M2} up.

Method I: the voltage as an unknown



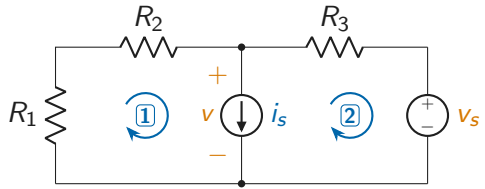
$$\begin{bmatrix} R_1+R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

The voltage v of i_s is one more unknown.

The constraint: i_s carries i_{M1} down and i_{M2} up.

Three unknowns, three equations; the last row is the constraint.

Method I: the voltage as an unknown



$$\begin{bmatrix} R_1+R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

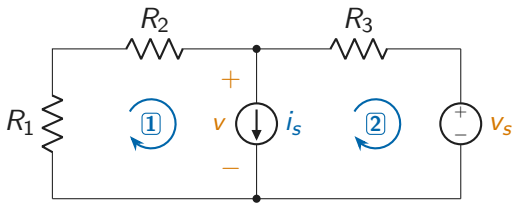
The voltage v of i_s is one more unknown.

The constraint: i_s carries i_{M1} down and i_{M2} up.

Three unknowns, three equations; the last row is the constraint.

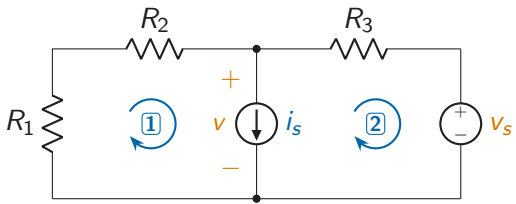
It works, but the supermesh avoids the extra unknown.

Method II: the supermesh



Question. Can we get rid of the unknown v?

Method II: the supermesh

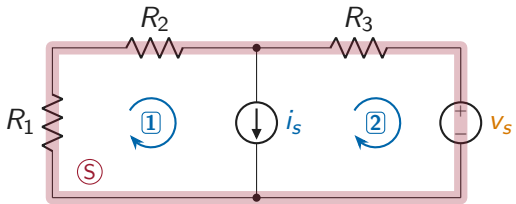


Question. Can we get rid of the unknown v?

Add the KVL equations of meshes 1 and 2.

v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels.

Method II: the supermesh



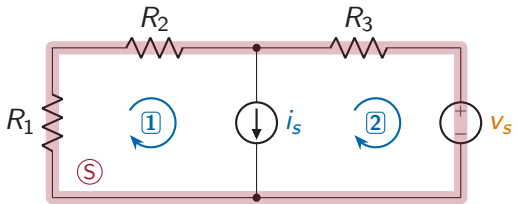
Question. Can we get rid of the unknown v?

Add the KVL equations of meshes 1 and 2.

v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels.

This is KVL around the outer loop Ⓢ of the two meshes; it avoids i_s.

Method II: the supermesh



Question. Can we get rid of the unknown v?

Add the KVL equations of meshes 1 and 2.

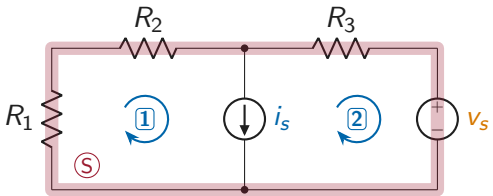
v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels.

This is KVL around the outer loop Ⓢ of the two meshes; it avoids i_s.

Supermesh

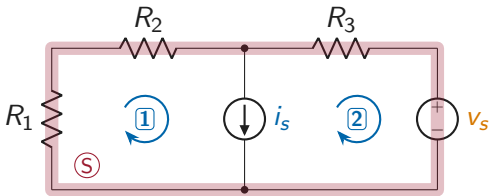
Write KVL around two meshes as if they were a single mesh, skipping the current source between them.

Method II: the equations



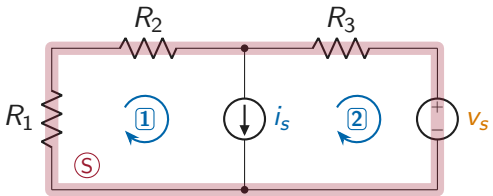
Question. Write KVL around the supermesh.

Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

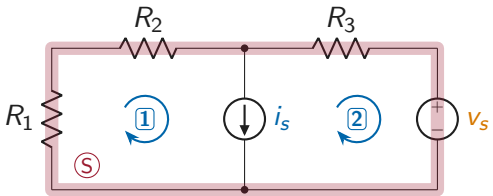
Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Method II: the equations

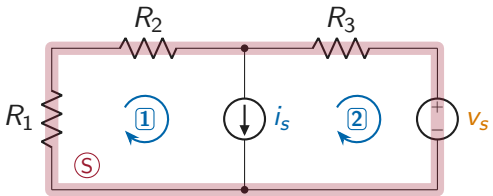


Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

Method II: the equations



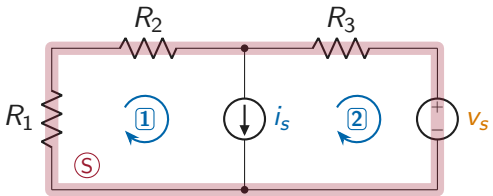
Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

Two unknowns: one fewer than Method I.

Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

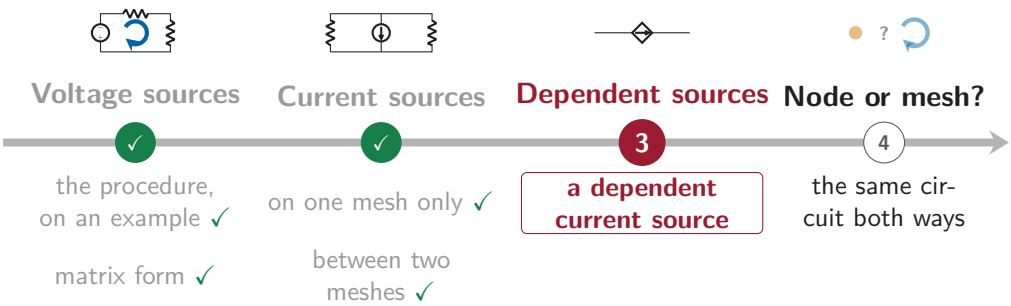
Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

Two unknowns: one fewer than Method I.

The matrix is not symmetric any more.

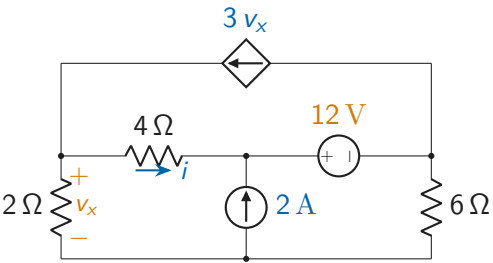
Roadmap



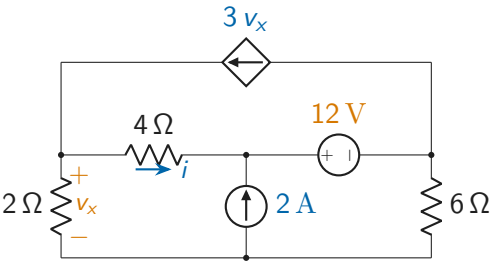
Each station adds a new kind of source; the last one compares the two methods.

Example 12.4: a dependent current source

Question. Find i, the current of the $4\,\Omega$.



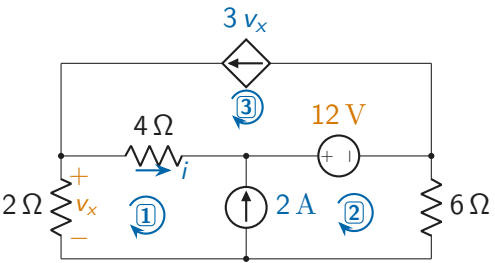
Example 12.4: a dependent current source



Question. Find i, the current of the 4 Ω.

Treat the dependent source as a current source of value $3\,v_x$.

Example 12.4: a dependent current source

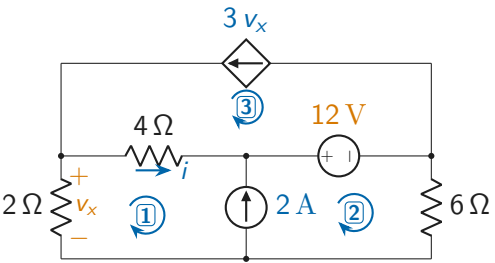


Question. Find i, the current of the 4 Ω.

Treat the dependent source as a current source of value $3\,v_x$.

Three inner meshes: unknowns i_{M1}, i_{M2}, i_{M3}.

Example 12.4: a dependent current source



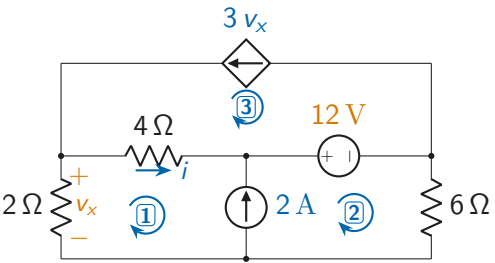
Question. Find i, the current of the 4 Ω.

Treat the dependent source as a current source of value $3\,v_x$.

Three inner meshes: unknowns i_{M1}, i_{M2}, i_{M3}.

The 2 A source is between meshes 1 and 2: a supermesh.

Example 12.4: a dependent current source



Question. Find i, the current of the 4 Ω.

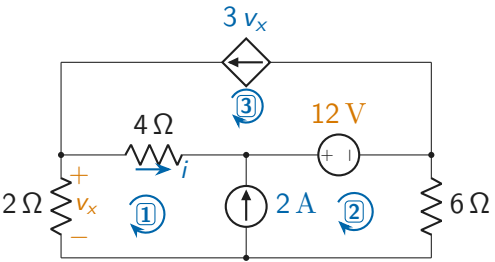
Treat the dependent source as a current source of value $3\,v_x$.

Three inner meshes: unknowns i_{M1}, i_{M2}, i_{M3}.

The 2 A source is between meshes 1 and 2: a supermesh.

The dependent source is on mesh 3 only: it fixes i_{M3}.

Example 12.4: a dependent current source



Question. Find i, the current of the 4 Ω.

Treat the dependent source as a current source of value $3\,v_x$.

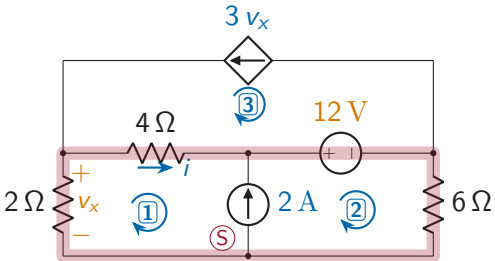
Three inner meshes: unknowns i_{M1}, i_{M2}, i_{M3}.

The 2 A source is between meshes 1 and 2: a supermesh.

The dependent source is on mesh 3 only: it fixes i_{M3}.

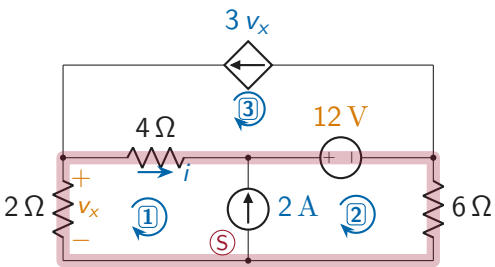
Equations: KVL around the supermesh and two constraints.

Example 12.4: the equations



Question. Write KVL around the supermesh.

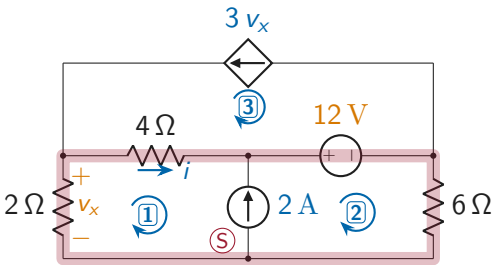
Example 12.4: the equations



Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

Example 12.4: the equations

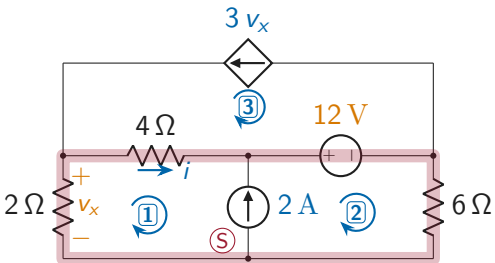


Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

The $2\,\mathrm{A}$ source: $i_{M2} - i_{M1} = 2 \quad (2)$

Example 12.4: the equations



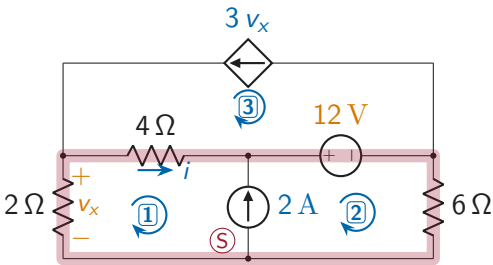
Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

The $2\,\mathrm{A}$ source: $i_{M2} - i_{M1} = 2$ (2)

The dependent source: $i_{M3} = -3\,v_x$

Example 12.4: the equations



Supermesh:

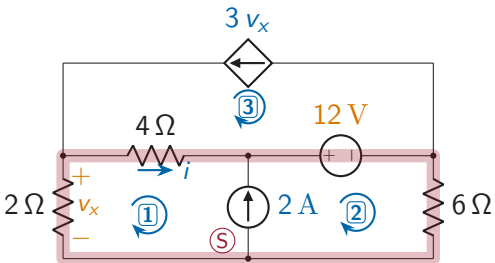
$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

The $2\,\mathrm{A}$ source: $i_{M2} - i_{M1} = 2$ (2)

The dependent source: $i_{M3} = -3\,v_x$

Question. What is v_x in mesh currents?

Example 12.4: the equations



Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

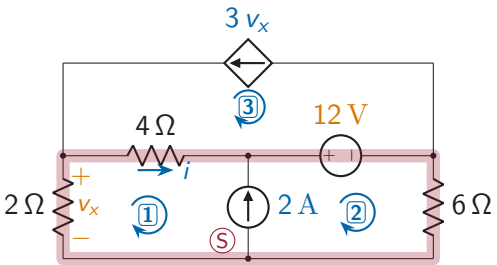
The $2\,\mathrm{A}$ source: $i_{M2} - i_{M1} = 2$ (2)

The dependent source: $i_{M3} = -3\,v_x$

Question. What is v_x in mesh currents?

i_{M1} flows up through the $2\,\Omega$, from $-$ to $+$: $v_x = -2\,i_{M1}$.

Example 12.4: the equations



Supermesh:

$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$

The 2 A source: $i_{M2} - i_{M1} = 2 \quad (2)$

The dependent source: $i_{M3} = -3\,v_x$

Question. What is v_x in mesh currents?

i_{M1} flows up through the 2 Ω, from − to +: $v_x = -2\,i_{M1}$.

Constraint: $i_{M3} = 6\,i_{M1} \quad (3)$

Example 12.4: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

Example 12.4: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

Rows: KVL around the supermesh, the $2\,\mathrm{A}$ source, the dependent source.

Example 12.4: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

Rows: KVL around the supermesh, the $2\,\mathrm{A}$ source, the dependent source.

Solve: $i_{M1} = 2\,\mathrm{A}$, $i_{M2} = 4\,\mathrm{A}$, $i_{M3} = 12\,\mathrm{A}$.

Example 12.4: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

Rows: KVL around the supermesh, the $2\,\mathrm{A}$ source, the dependent source.

Solve: $i_{M1} = 2\,\mathrm{A}$, $i_{M2} = 4\,\mathrm{A}$, $i_{M3} = 12\,\mathrm{A}$.

$$i = i_{M1} - i_{M3} = 2 - 12 = -10\,\mathrm{A}$$

Dependent sources

- A dependent source is handled like an independent one of the same kind.

Dependent sources

- A dependent source is handled like an independent one of the same kind.
- Its control variable (v_x here) is written with mesh currents: this gives the constraint equation.

Dependent sources

- A dependent source is handled like an independent one of the same kind.
- Its control variable (v_x here) is written with mesh currents: this gives the constraint equation.
- A dependent voltage source is the easy case: it adds a term to KVL, just as a dependent current source added a term to KCL previously.

Dependent sources

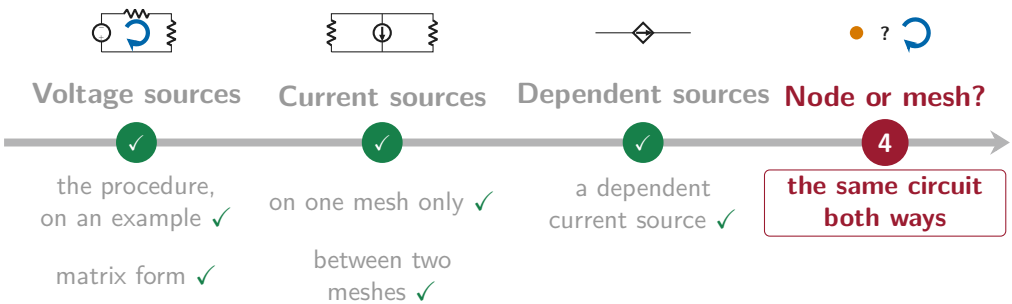
- A dependent source is handled like an independent one of the same kind.
- Its control variable (v_x here) is written with mesh currents: this gives the constraint equation.
- A dependent voltage source is the easy case: it adds a term to KVL, just as a dependent current source added a term to KCL previously.

Remark

If a dependent current source is on mesh k only, do not write KVL around mesh k: its voltage is not known.

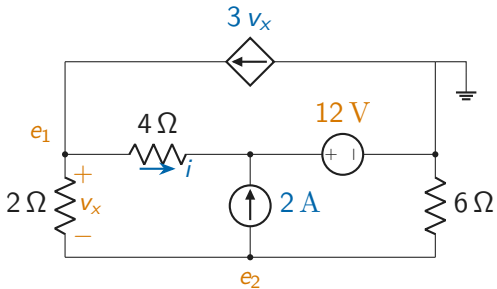
Write the constraint that the dependent source gives instead.

Roadmap



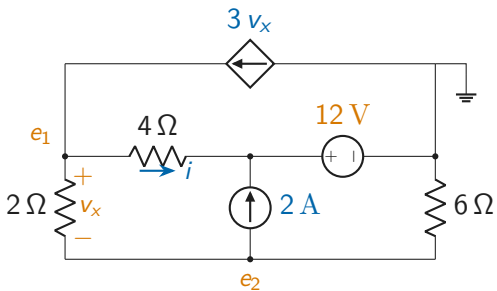
Each station adds a new kind of source; the last one compares the two methods.

Example 12.4 with node voltages



Question. Solve the same circuit with node voltages.

Example 12.4 with node voltages



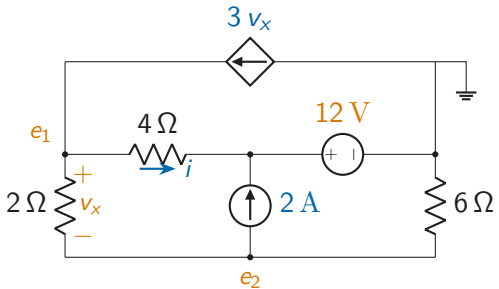
Question. Solve the same circuit with node voltages.

Take the right node as the ground.

The 12 V source goes from the middle node to the ground: that node is at 12 V.

Unknowns: e_1, e_2.

Example 12.4 with node voltages



Question. Solve the same circuit with node voltages.

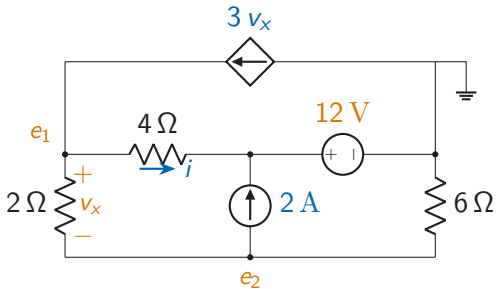
Take the right node as the ground.

The 12 V source goes from the middle node to the ground: that node is at 12 V.

Unknowns: e_1, e_2.

Node 1: $\dfrac{e_1 - e_2}{2} + \dfrac{e_1 - 12}{4} - 3\,v_x = 0$

Example 12.4 with node voltages



Question. Solve the same circuit with node voltages.

Take the right node as the ground.

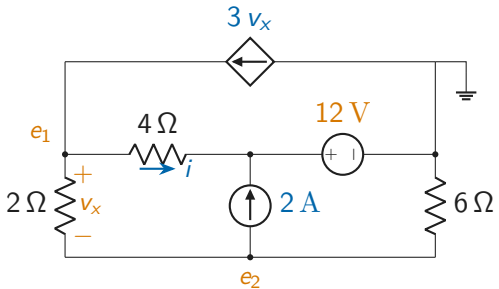
The 12 V source goes from the middle node to the ground: that node is at 12 V.

Unknowns: e_1, e_2.

Node 1: $\dfrac{e_1 - e_2}{2} + \dfrac{e_1 - 12}{4} - 3\,v_x = 0$

Node 2: $2 + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{6} = 0$

Example 12.4 with node voltages



Question. Solve the same circuit with node voltages.

Take the right node as the ground.

The 12 V source goes from the middle node to the ground: that node is at 12 V.

Unknowns: e_1, e_2.

Node 1: $\dfrac{e_1 - e_2}{2} + \dfrac{e_1 - 12}{4} - 3\,v_x = 0$

Node 2: $2 + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{6} = 0$

Constraint: $v_x = e_1 - e_2$

Example 12.4 with node voltages: matrix form

Put $v_x = e_1 - e_2$ and collect the terms:

$$\begin{bmatrix} -\frac{9}{4} & \frac{5}{2} \\ -\frac{1}{2} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

Example 12.4 with node voltages: matrix form

Put $v_x = e_1 - e_2$ and collect the terms:

$$\begin{bmatrix} -\frac{9}{4} & \frac{5}{2} \\ -\frac{1}{2} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \qquad \Rightarrow \qquad e_1 = -28\,\mathrm{V}, \quad e_2 = -24\,\mathrm{V}$$

Example 12.4 with node voltages: matrix form

Put $v_x = e_1 - e_2$ and collect the terms:

$$\begin{bmatrix} -\frac{9}{4} & \frac{5}{2} \\ -\frac{1}{2} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \qquad \Rightarrow \qquad e_1 = -28\,\mathrm{V}, \quad e_2 = -24\,\mathrm{V}$$

$$i = \frac{e_1 - 12}{4} = \frac{-28 - 12}{4} = -10\,\mathrm{A}$$

Example 12.4 with node voltages: matrix form

Put $v_x = e_1 - e_2$ and collect the terms:

$$\begin{bmatrix} -\frac{9}{4} & \frac{5}{2} \\ -\frac{1}{2} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \qquad \Rightarrow \qquad e_1 = -28\,\mathrm{V}, \quad e_2 = -24\,\mathrm{V}$$

$$i = \frac{e_1 - 12}{4} = \frac{-28 - 12}{4} = -10\,\mathrm{A}$$

The same answer, as it must be.

Node or mesh?

Question. For Example 12.4, which method needs fewer equations?

Node or mesh?

Question. For Example 12.4, which method needs fewer equations?

Mesh currents

$\ell = 3$ unknowns.

Each current source fixes one of them:
$3 - 2 = 1$ KVL equation is left.

Node or mesh?

Question. For Example 12.4, which method needs fewer equations?

Mesh currents

$\ell = 3$ unknowns.

Each current source fixes one of them:
$3 - 2 = 1$ KVL equation is left.

Node voltages

$n - 1 = 3$ unknowns.

Each voltage source fixes one of them:
$3 - 1 = 2$ KCL equations are left.

Node or mesh?

Question. For Example 12.4, which method needs fewer equations?

Mesh currents

$\ell = 3$ unknowns.

Each current source fixes one of them:
$3 - 2 = 1$ KVL equation is left.

Node voltages

$n - 1 = 3$ unknowns.

Each voltage source fixes one of them:
$3 - 1 = 2$ KCL equations are left.

Tip. Count before you start: mesh, ℓ minus the current sources; node, $n - 1$ minus the voltage sources.

Mesh current analysis needs a planar circuit; node voltage analysis always works.

Summary

In the circuit	What to do
resistors and voltage sources	KVL around every inner mesh: $Z_M \mathbf{i}_M = \mathbf{v}_M$, from three simple rules; $Z_M = M_R R M_R^T$ is symmetric
a current source on one mesh only	that mesh current is known; no KVL around that mesh
a current source between two meshes	Method I: its voltage as an unknown, plus the constraint; Method II: KVL around the supermesh, plus the constraint
a dependent source	write its control variable with mesh currents: a constraint equation
node or mesh?	count the equations; mesh needs a planar circuit

After the mesh currents are found: $\mathbf{i} = M^T \mathbf{i}_M$, then the branch voltages from the element laws.